User Tip: *When using grade sensors with detachable temperature bails, never use a sensor without the temperature bail. Be sure the temperature bail is in good condition and attaches firmly to the grade sensor. It is a good idea to have some spare temperature bails in the storage box for the grade sensors.*

Material spilling over the hopper sides is a common way that the target area for the grade sensor is contaminated. The grade sensor is an extremely sensitive measuring device and will detect even small aggregate particles. Find a way to prevent spills. Do not continue to operate with grade control unless a suitable grade target is consistently available.

Other things that may interfere with the sensor target area include laborers who are working around the paver, debris that may be knocked from overhanging trees, and spills from haul units as they pull away from the paver.

The height of the grade sensors above the grade reference should be checked while setting up the grade control system. The farther the grade sensor is from the reference, the larger the target area becomes.



Spills over the hopper sides may end up in the target area of the grade sensor.

Cat grade sensors are functional from 20-100 cm (8-39") above the target grade reference. If the distance from the target is outside that range, the grade control system will not function. Caterpillar recommends positioning grade sensors, so they are 40-46 cm (16-18") above the target.

Note: The functional range of other types of grade sensors may be different. Always follow manufacturers' guidelines. However, the 40-46 cm (16-18") height recommendation is usually applicable to other types of grade sensors.



The recommended height above the grade reference is 40 to 46 cm (16-18").

User Tip: *When a sonic averaging ski is installed, multiple grade sensors will be on the ski. Try to set all sensors within the recommended height range. The sensors do not have to be exactly the same height. However, try to install the ski, so it is parallel to the grade and the sensors do not have significant height differential.*

Caterpillar also recommends that the grade sensors be positioned as close to the edge of the paving width as possible. The location of the sensor becomes the control point for all thickness changes. It is especially important to position the grade sensor close to the paving width, if one side of the screed is under grade control and the other side is under slope control. The sensor can be positioned just outside the paving width or just inside the paving width, depending on the paving application and the suitability of the grade reference.

[MECHANICAL GRADE SENSORS]

Mechanical grade sensors, also referred to as contact-type sensors, touch the grade reference. Mechanical grade sensors are not subject to variations related to air temperature changes. Mechanical sensors are often installed on pavers that vent the engine cooling system on the side of the paver or when there are very windy conditions.

A mechanical sensor uses a rotary sensor. An arm extends from the rotary sensor. The arm can have a matching shoe that rides on the grade that is the reference or the arm can have a wand that contacts a stringline. The arm should run at about a 45 degree angle, always trailing behind the sensor. As the matching shoe or wand traces deviations in the grade reference, the rotary sensor detects changes in the angle of the trailing arm. The changes in the angle of the sensor arm are converted to distances and these measurements are sent to the control panel or control module.



Cat contact-type grade sensor equipped with joint-matching shoe.

The pressure on the wand is adjustable. Set the arm pressure so the wand stays in contact with the string, but does not deflect the string. The Cat rotary contact sensor can be set up to trace the top of the string or can be set up to ride underneath the string.



The mechanical sensor can be equipped with a wand that traces a stringline over an averaging ski.

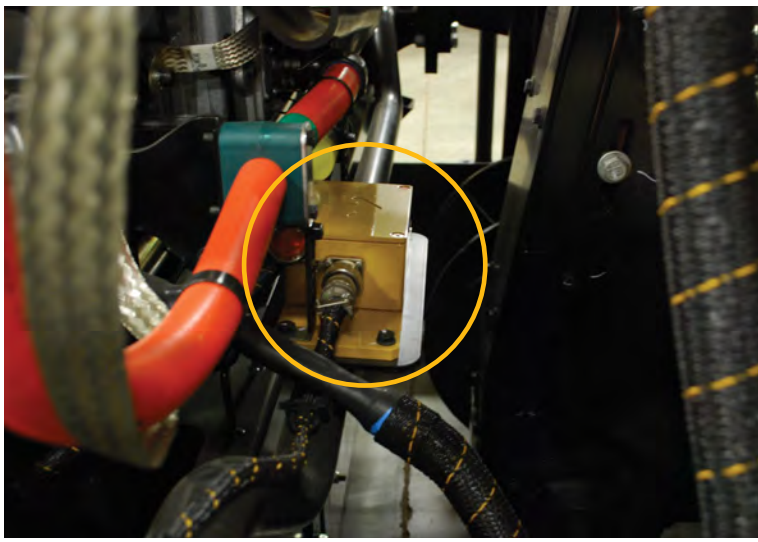
[SLOPE SENSOR]

The slope sensor is installed on a transverse beam that spans the width of the screed directly above the front of the main screed.

The slope sensor functions like a precision carpenter's level. It is a sealed sensor that remains on the paver and does not need to be removed and stored at the end of the shift. The angle sensor reads the slope of the screed and sends that reading to the control panel or control module. The slope sensor allows the crew to maintain slope from right to left or from left to right. Slope is measured in percent. Up to 10 percent positive or negative slope is possible when using automatic slope control.

The slope sensor is connected to the left and right control panels or display units. Only one side of the screed can be under slope control at one time.

Calibrate electronic slope sensors by using a mechanical or electronic level to measure the slope of the grade that the screed is resting on. Then, edit the slope value displayed on the control panel or display unit.



The Cat slope sensor is an electronic slope sensor.

Note: *Some systems still use hydraulic angle sensors with a pendulum suspended in oil. Calibrate hydraulic sensors according to the manufacturer's guidelines.*

[DISPLAY UNITS AND CONTROL PANELS]

Cat Pavers can be equipped with different types of grade and slope control systems. Some systems use control panels that combine displays and control modules into one device. The Cat Grade Control system has separate displays and a separate control module. Most of the information about system operation and many of the visuals in this unit cover the Cat Grade system on Cat F-Series Pavers.



Cat Grade Control on F-Series Paver.



Note: *Some Cat Pavers have grade and slope control that is installed at the factory. Those pavers use the screed control module for the grade and slope system. Other Cat Pavers have grade and slope systems that are installed by Cat dealers or by equipment users. Those pavers have a dedicated control module for grade and slope control. Always read and follow the guidelines contained in each grade and slope system's Operation and Maintenance Manual.*



Home Menu screen on the Cat SE-Series screed control box display.

When the key switch is in the ON position, the displays at the operator's stations and on both sides of the screed will illuminate and show the Home Menu screen.

An operator can select the grade and slope screen by depressing the button adjacent to the grade / slope icon at the upper left of the display housing.

The screed operator(s) can elect to work with single screen display or with dual screen display.

grade control menu



Grade Control display showing single screen mode for right side of the screed.



In the Operator Settings menu, an operator can set single screen or dual screen display.

To select the display mode touch the grade control menu and select operator settings. Select main screen and touch the 2D guidance screen mode.

After the change to the display screen mode is accepted, select the grade control icon (lower right) to return to the normal operating screen.



Dual screen mode showing elevation (grade) control right side and slope control left side

This illustration shows the Cat Grade Control display set in single screen mode on the right side of the screed. Functional areas are grouped for ease of understanding.



1. Grade indicators are on the far right side of the display. The green bars illuminate when the grade or slope system is within the dead band. This condition is usually referred to as being "on grade." The up and down arrows flash when the measured grade or slope value is outside of the dead band.

2 Adjustment and selection buttons are also on the right side of the display. Depressing the buttons adjacent to the up-and-down arrows enable the system to make grade or slope changes while paving in the Auto mode.

3. Set the benchmark (null) by depressing button adjacent to the bench key.

4. Switch between the Auto mode and the Manual mode by depressing the button adjacent to the Auto symbol.

5. The Grade Control menu is located in the upper right portion of the screen. Select this touch-screen button to change units of measure, control system beepers, and other preferences for the grade control system.

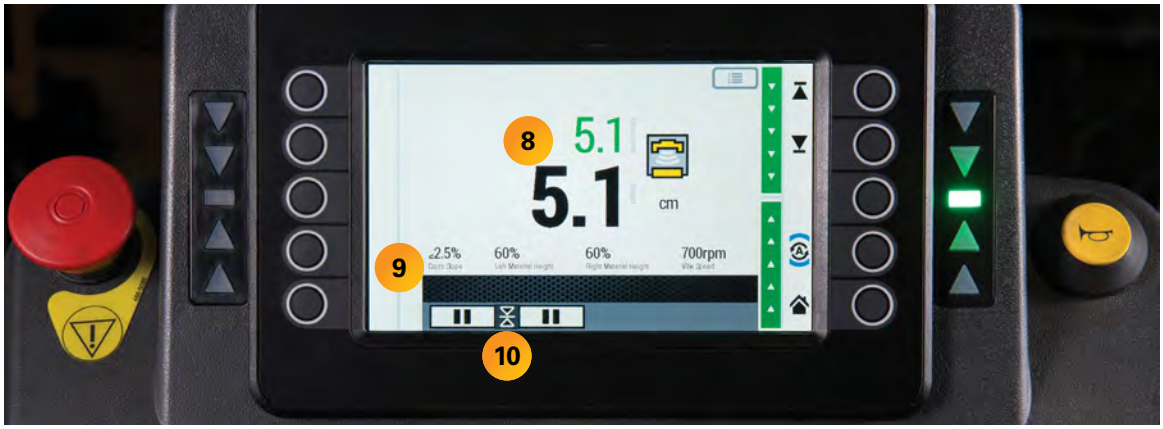
6. The Sensor Selection icon is in the right center of the screen. Select this touch-screen icon to see what sensors are available and to select a different sensor.

7. Finally, depress the button adjacent to the Home icon to return to the Home screen

Once the sub-menu for Sensor Selection displays, select the icon for the sensor to be used. The sensor will highlight. Depress the button adjacent to the OK symbol to accept the change.



Touching the Sensor Selection icon opens the sub-menu of available sensors.



8. The measured reference number (smaller green) and the target reference number (larger black) are in the center of the screen. The measured number is the value sent to the display by the selected sensor. While the paver is moving, the number will usually be changing slightly as the measured grade or measured slope changes. The target reference number is a value selected by the operator and usually refers to the desired thickness or desired slope. The reference number stays the same unless the operator uses the up or down arrows to change thickness or slope. Edit the target number by tapping the touch screen value. Use the sub-menu to change the target value and move the system to a new position or target value.

9. The operator can choose to include a text bar at the bottom of the screen. A number of values are available for viewing. For example, the operator may wish to monitor certain feeder system functions, while using the grade control screen. Or, the operator may wish to view the slope of the screed, while operating in grade control. To use the text bar, select the grade control menu. Select operator settings and text slider bar to "customize" information that is available to view.

10. The bottom section of the display indicates the state of the system. When the "hand" symbol shows in this area, the control system is in the Manual mode. When the word "AUTO" shows in this area, the system is in the Automatic mode. When the Pause symbol (two vertical lines) shows in this area, the system is momentarily paused from the Auto mode. The pause symbol will appear when the operator moves the propel lever to the Neutral position, for example.

Selecting Type of Sensor



Dual screen display with single sensor grade control left and slope control right.

To select the type of sensor or averaging ski set-up, tap the sensor selection icon on the touch-screen display. The sub-menu for sensor selection will show each sensor or ski configuration that is available.

sonic over a string line

sonic over a hard surface

sonic in center of averaging beam

sonic in front and rear of averaging beam

all 3 sensors on averaging beam

contact sensor

cross slope sensor

3D slope sensor

3D elevation sensor

Select the icon for the sensor or configuration to be used. Depress the button adjacent to the OK symbol and then return to the desired display screen. The selected sensor icon(s) will appear on the Grade Control display.

When a sonic sensor over a string line is selected, the icon on the display guides the paver operator to keep the sensor centered over the stringline. The Cat grade sensor uses the center three transducers to measure the distance to the stringline. The Cat grade sensor uses the outer two transducers as centering guides. If the sensor is centered over the string, there will be a green circle around the string on the icon. If the sensor is offset to the left of center, there will be a red arrow pointing to the right. If the sensor is offset to the right, there will be a red arrow pointing to the left.



sonic tracer centered over string line



sonic tracer offset to the left of the string line



sonic tracer offset to the right of the string line

User Tip: *If the grade sensor is set up to measure a string over the body of a mechanical averaging ski as its grade reference, position the string 10 cm (4") above the body of the ski. If the grade sensor moves away from the target string, the grade control system will lock to grade and the tow-point will not move. The lock-to-grade feature activates anytime there is a sudden change in the grade reference in excess of 5 cm (2").*

Note: *When using a sonic grade sensor over a stringline and an icon for stringline guidance is not displayed, then the grade sensor is not tracking the stringline. For proper guidance, the sensor connector must be on the left side when viewing from the back of the paver.*

Benchmarking or Nulling the System

Once the paver and screed have been setup to pave using "Paving by the Numbers" and the Grade Control sensor/sensors have been installed, the system can be bench-marked or nulled.

Establishing a benchmark (nulling the system) creates a reference for the grade sensor or for the slope sensor. In the case of a grade sensor, the benchmark is the distance from the sensor to the grade reference. In the case of a slope sensor, the benchmark is the percentage of slope from right to left or from left to right.

To benchmark the system, depress and hold the left or right Bench button (middle button) for longer then two seconds. If the audio function is activated, the operator will hear three beeps, and the middle on-grade bar will illuminate as a green color.

When benchmarking the system under grade control, the value in the Measured value area is converted to the number shown in the Target value area. Benchmarking the system under slope control will convert the measured value to the target value.



The green on-grade bar illuminates when the system is benchmarked.

Selecting Automatic Mode

After benchmarking the system, depress and release the button adjacent to the AUTO symbol, the Mode Select button (second from bottom). The word "AUTO" will appear in the system status area. While the paver is placing material, AUTO will show continuously in the system status area.

When the paver is stopped, the grade and slope control system is temporarily in the Standby mode and the Pause symbol will alternate with the word Auto in the system status area. When the propel lever is moved out of the Neutral position, the system returns to the Auto mode and the word Auto will appear constantly in the system status area.

When paving in automatic mode, the control module will send signals to the tow-point valve. When a change in elevation or slope is required, the left or right tow-point valve will send hydraulic oil to the head end or rod end of the appropriate tow-point cylinder. The tow-point will move up or down to change the elevation or the slope to within the tolerance for grade and slope. The grade control system has a default tolerance (dead band) of 3 mm (0.125"). The slope control system has a default tolerance of 0.05%. Grade and slope tolerances can be adjusted, if desired.



The word "AUTO" appears when the automatic mode is selected.



The Pause symbol will alternate with the word Auto when the propel lever is in the Neutral position.

Editing the Measured Value

The value in the measured area is a reference for the operator. It is a good practice to edit that value, so it corresponds with the desired thickness of the bituminous layer or with the specified slope of the bituminous layer. To edit this value, select the measured value on the touch-screen display.

When the measured value is selected, a keypad will display. The desired value is entered on the keypad, either for grade sensor distance (elevation control) or for slope percentage.



A keypad displays when the measured value is selected.

The value entered will display at the top of the keypad. The value will be expressed in the selected unit of measure for the grade control system. To accept the change, depress the button adjacent to the OK symbol.

After returning to the grade control display, bench the system by depressing the left or right benchmark button (center button) for more than two seconds. Three beeps will be heard, and the target value will convert to the measured value that was just entered.



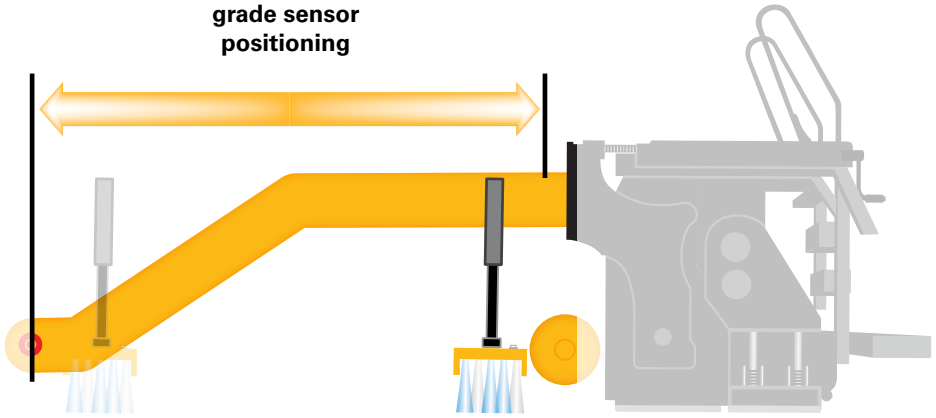
Keypad showing 2.5 inches entered as the measured value.



Bench the system and the target value will match the measured value.

[HOW SENSOR POSITION AFFECTS SCREED REACTION]

The position of a single grade sensor in relation to the tow-point connection or to the auger shaft affects how quickly the screed reacts to changes or corrections made by automatic grade control. Remember, when automatic grade and slope control is being used, the system automatically makes grade or slope changes by sending hydraulic oil through the tow-point valves to the tow-point cylinders. How far the tow-points move up and down determines the amount that the main screed angle of attack changes and, consequently how far and how fast the screed changes its elevation. Operators must understand these relationships in order to set up the paver properly for different paving applications.



Never position a single sensor in front of the tow-point connection.

A single grade sensor may be positioned anywhere between the tow-point connection and the center of the auger shaft when paving with a vibratory screed.

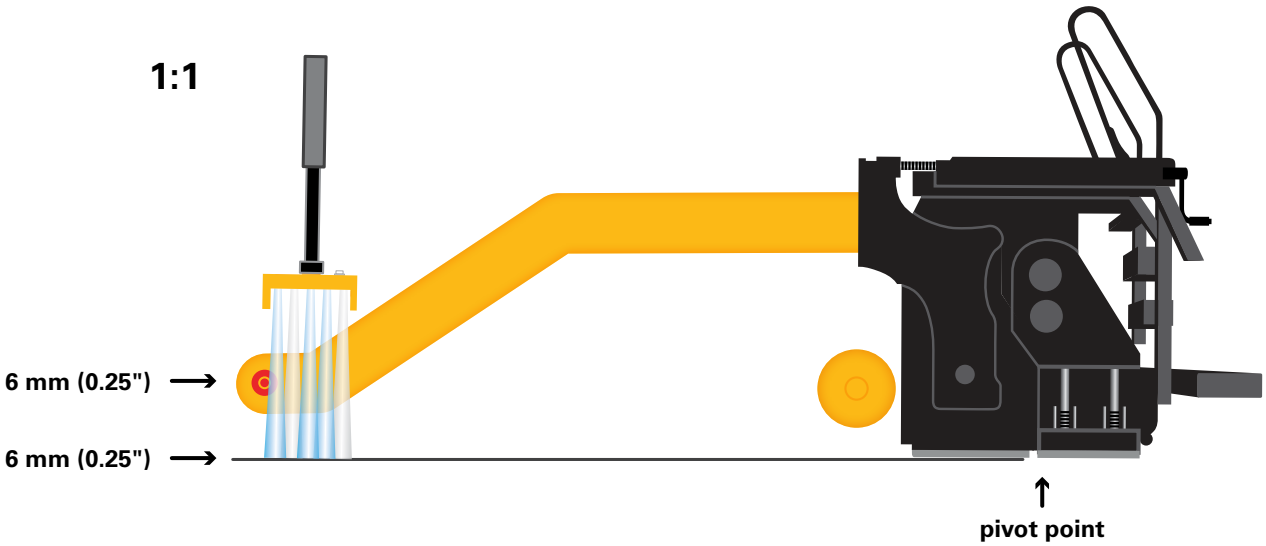
Note: *When paving with a tamping screed, a sensor may be positioned as far back as even with the tamper bar.*

Assume a single grade sensor is installed in front of the tow-point and the sensor detects a change in the grade elevation. If the change is a depression in the grade, the control module will signal the tow-point cylinder to move up. Conversely, if the grade sensor detects a high spot in the grade, the control module will signal the tow-point cylinder to move down. In other words, all grade corrections are the opposite of what is required when the grade sensor is ahead of the tow-point connection.

User Tip: *The Cat Grade Control System recognizes when a non-contact averaging ski is installed on the side of the paver. There are three sensors on the ski and, when three-sensor or two-sensor operation is shown in the Sensor Selection area, the system averages the measurements from the sensors. Don't worry that one of the sensors is in front of the tow-point connection. The system recognizes and accounts for that condition. However, if one-sensor operation of the non-contact averaging ski is selected, make sure that the middle sensor is behind the tow-point connection. The beam may have to be moved toward the rear to place the middle grade sensor in an acceptable position.*

Next, assume that a single sensor is behind the auger shaft or behind the tamper bar, and the sensor detects a change in grade elevation. By the time the system responds, it will probably be too late. The change in elevation will occur after the screed has passed the point where the deviation is located, and extreme roughness will be evident in the mat.

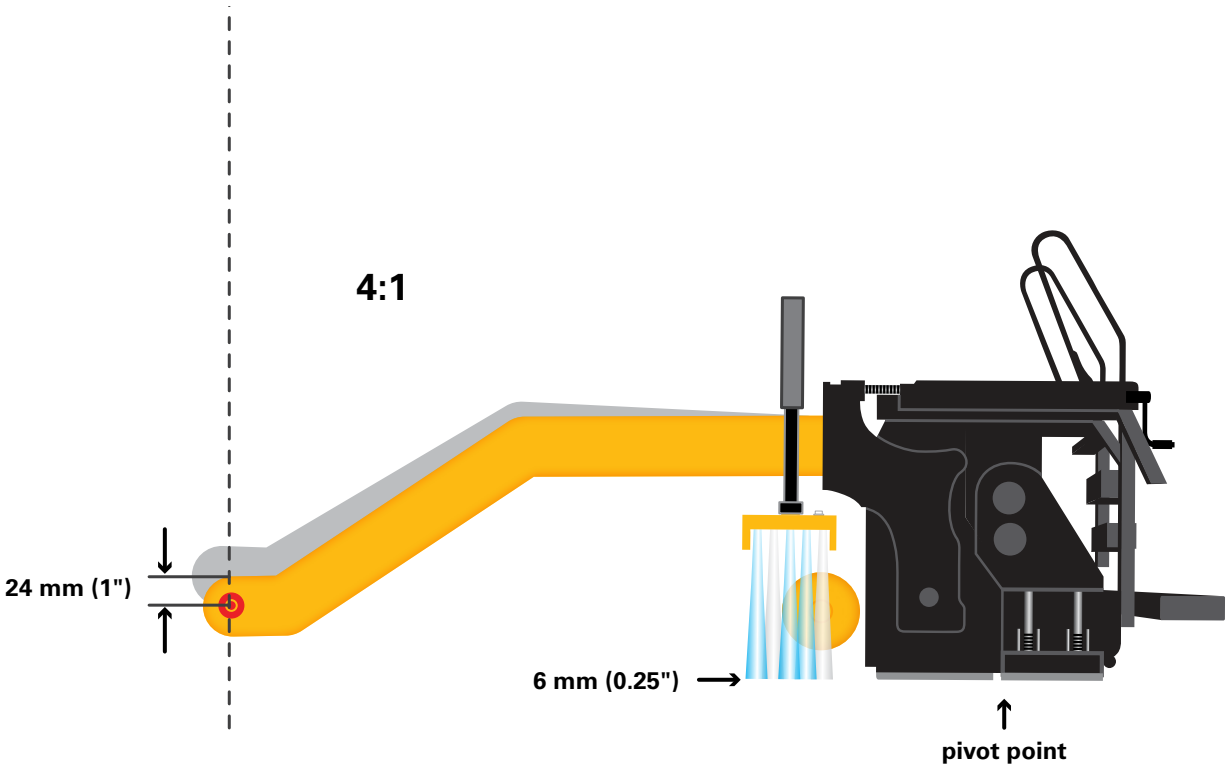
Always verify that the sensor is located correctly and that it will create the desired screed response.



Sensor position for slow screed reaction.

To create a screed that reacts slowly to grade changes, position the grade sensor just behind the tow-point connection. This sensor position creates a 1:1 ratio between the amount of grade deviation and the distance the tow-point moves. For example, if the sensor in this forward position detects a grade deviation that is 6 mm (0.25"), the grade control system will cause the tow-point to move 6 mm (0.25").

The main screed angle of attack changes slightly and the screed will climb or fall slowly. The change in elevation will occur gradually over a distance of approximately five tow-arm lengths. The bituminous layer will tend to fill in low spots in the grade and trim off high spots on the grade with the sensor close to the tow-point. Select this sensor position when it is important to improve the ride quality of the grade.



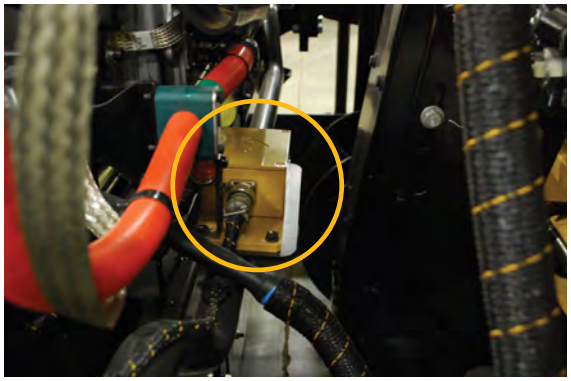
Sensor position for fast screed reaction.

To create a screed that reacts quickly to grade changes, position the grade sensor in line with the center of the auger shaft or even with the tamper bar. This sensor position creates a 4:1 ratio between the amount of grade deviation and the distance the tow-point moves. For example, if the sensor in this position detects a grade deviation that is 6 mm (0.25"), the grade control system will cause the tow-point to move 24 mm (1").

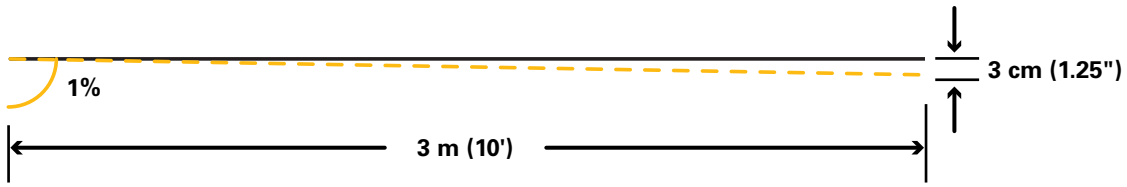
The main screed angle of attack changes significantly and the screed will climb or fall quickly. The change in elevation will occur almost instantly, and the bituminous layer will exactly match the change in elevation where the change occurs. The bituminous layer will tend to follow low spots or high spots in the grade with the sensor close to the auger shaft. The thickness of the layer will be very uniform, but smoothness may be sacrificed. Select this sensor position when it is important to have precise yield control or exact height match to a longitudinal joint or gutter.

Position the grade sensor ahead of the auger shaft or the tamper bar to begin to slow screed reaction. This will create a layer with an acceptable yield and profile while improving the smoothness at the same time.

The slope sensor is mounted on a transverse beam that spans the width of the main screed just ahead of the nose of the screed. In this position, the slope sensor also creates a 4:1 ratio between the distance that the tow-point moves in order to create the elevation change needed to maintain the specified slope. It is important for operators to understand how automatic slope control operates. First, the relationship between slope correction and elevation changes must be understood.

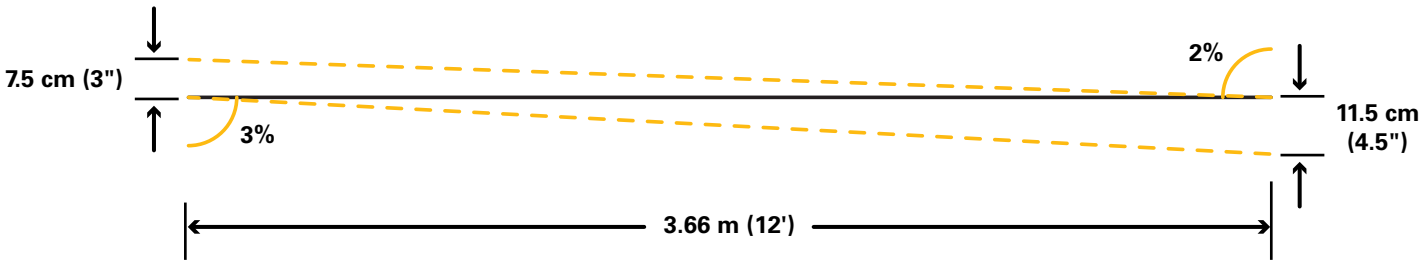


The position of the slope sensor also creates a fast screed reaction.

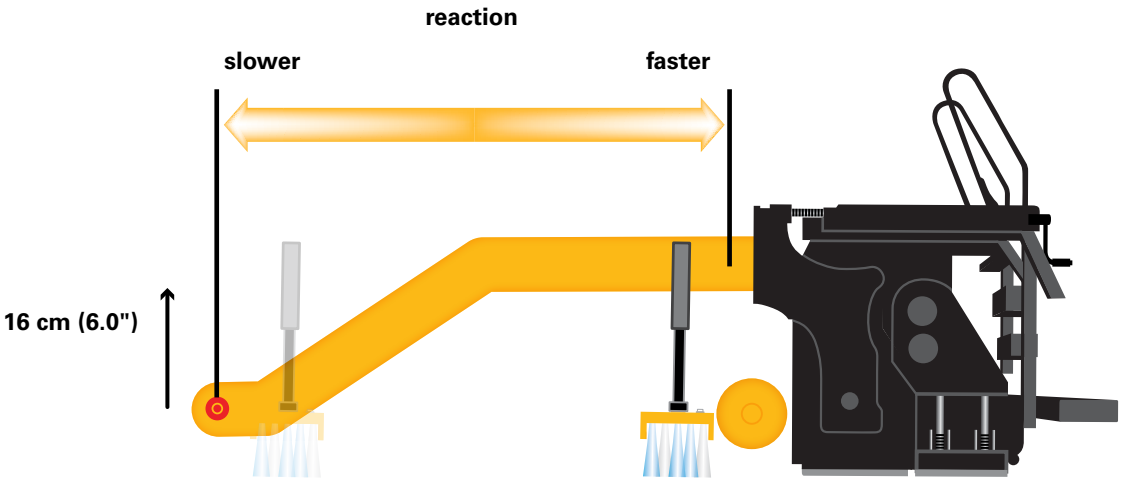


Assume that a structure needs to have a 1 percent slope from left to right. Starting at the left edge of the structure, the elevation of the layer must decrease by 1 cm per meter, or 0.125" per foot. If the width of the structure is 3 m (10'), then the elevation change will be 3 cm (1.25").

In practical terms, consider the following example. The slope of a compacted granular base is up to 3 percent from left to right in some areas. The plan specifies 2 percent slope. The paving crew needs to correct the slope to 2 percent in all areas, while paving the base layer. The width of the layer is 3.66 m (12'). The uncompacted depth of the layer is supposed to be 7.5 cm (3"). In those areas where the slope of the granular base is 3 percent, what will be the thickness of the layer at the right edge of paving?



In this example, the layer thickness has to increase by 4 cm (1.5") in order to create the specified slope. The crew will need to maintain the layer thickness at the left edge of paving. That is the only place where the operator can check layer thickness. The depth of the layer will increase as the layer extends from left to right. At the right edge, the layer thickness should be 11.5 cm (4.5") in order to correct the slope to 2 percent. How much will the right tow-point move in order to create the correct slope in this situation?



Because the slope sensor is located close to the front of the screed, the tow-point movement is a 4:1 ratio with the elevation change. Therefore, to create the 1 percent slope correction in a timely manner for this example, the tow-point will move up 16 cm (6").

Automatic slope control can create significant variations in layer thickness and, consequently, variations in yield. Automatic slope control can also create roughness in the bituminous layer.

User Tip: *When using automatic slope control to correct the transverse profile, it is a good idea to discuss the possible outcomes with the project owner and the on-site inspection personnel. Be sure to inform them that yields may be variable. If the inspection team is accustomed to checking yields truck by truck or over short distances, they may be surprised with the yield results, unless they have been prepared for the variations. It may be a good idea to use a slope board or slope meter to show the inspector where the corrections will occur.*

User Tip: *When using automatic slope control to correct the transverse profile, the demand for material passing under the screed will vary as the thickness of the layer varies. The paver and screed operators will have to be ready to adjust the feeder system frequently. They should pay particular attention to the area in front of the extender where the head of material will disappear the fastest. Auger speed will vary according to the demand for material, and stripe segregation may occur.*

[RESTORING TOW-POINT HEIGHT]

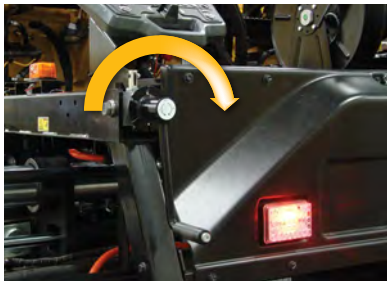
As discussed in previous sections of this guidebook, the height of the tow-points affects the line of pull and the main screed angle of attack. The tow-point height is set at the start of paving based on the planned thickness of the bituminous layer.

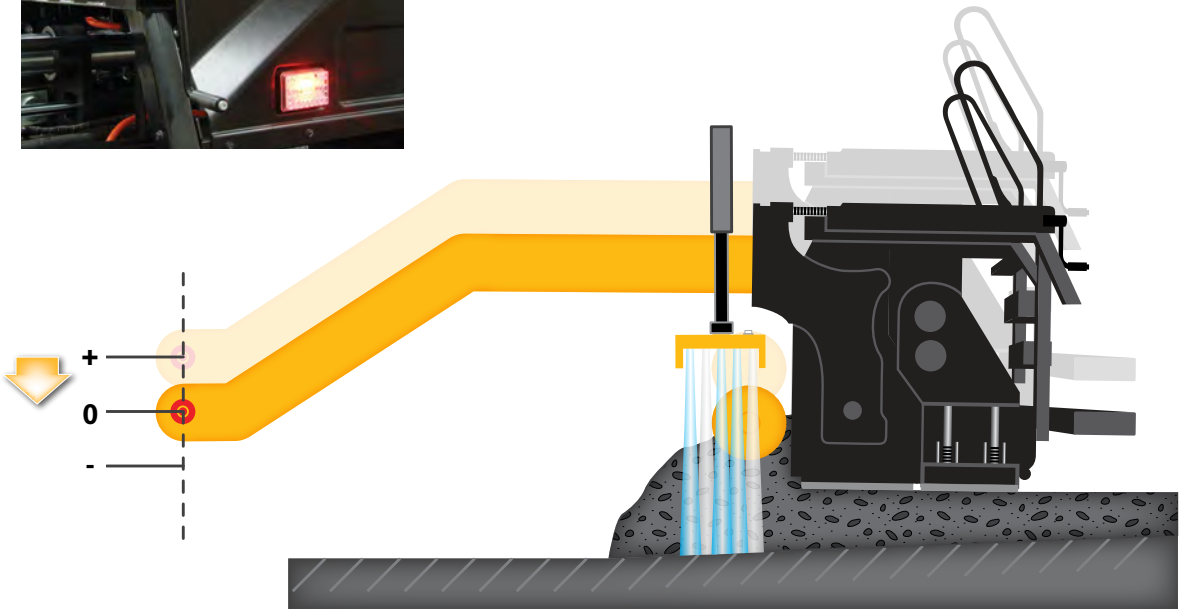


An operator can manually change tow-point height while paving in order to increase or decrease layer thickness. Or, an operator can use the grade and slope control system in the automatic mode to change tow-point height and layer thickness automatically.

The previous section described, how sensor position affects tow-point movement and screed reaction. Grade sensors positioned close to the nose of the screed can create large tow-point movements. Slope sensors, due to their fixed position close to the nose of the screed, can also create large tow-point movements.

Depending on the amount of grade deviation that is being corrected or the amount of slope correction, the tow-point cylinder may reach the end of its stroke, while the grade and slope control system is still signaling for additional movement. When that happens, the operator should employ a technique to "gain back" the tow-point cylinder to restore normal operation.



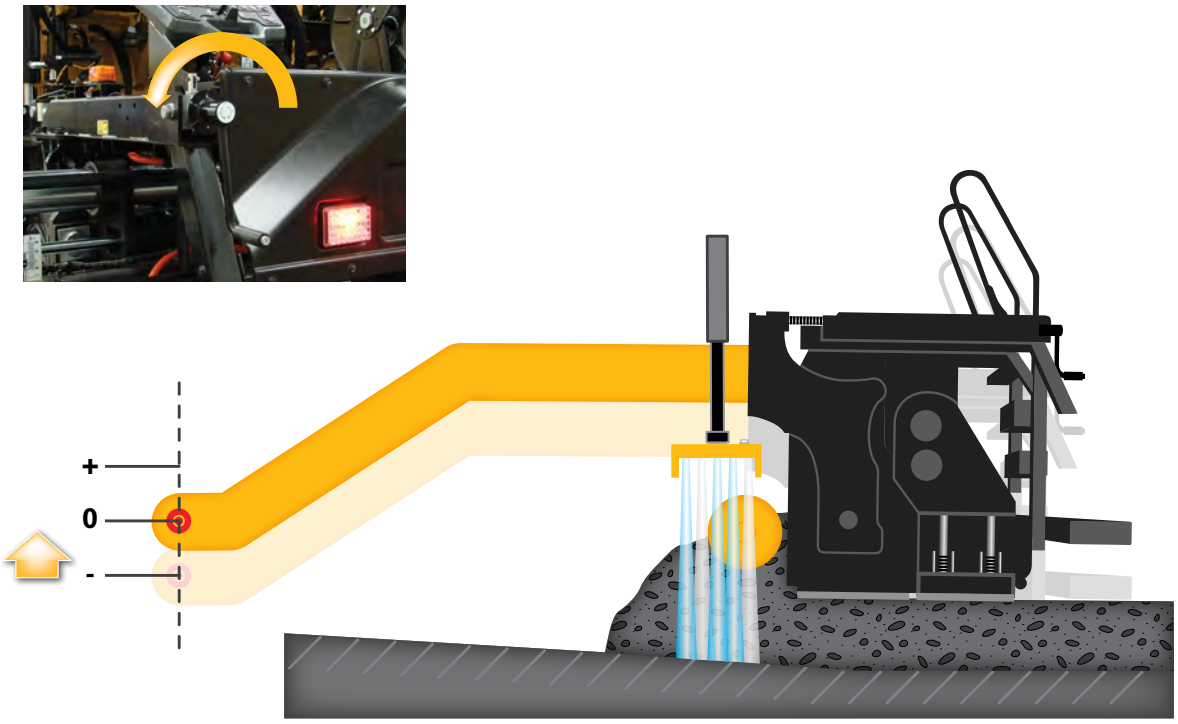


If additional thickness is required when the tow-point is fully raised and a single sensor Grade Control system is being utilized, Leave the system in the AUTO mode. On vibratory screeds, unlock the manual depth screw. Turn the screw slowly in the direction that increases layer thickness. Watch the tow-point height indicator. When the height indicator begins to move down, stop turning the manual depth screw. Allow the tow-point cylinder to stabilize. Make additional adjustments if needed. Once the desired tow-point position has been achieved, lock the depth control screw.

Note: *On tamper bar screeds, use the thickness screws to change the angle of attack on the main screed and start the tow-point height correction. Turn the screw toward positive marks to lower the tow-point. Turn the screw toward zero or negative marks to raise the tow-point.*

If the tow-point is too low, turn the manual depth screw or thickness screw in the direction that decreases layer thickness. Stop turning when the tow-point connection begins moving up. Allow the tow-point cylinder to stabilize. If additional adjustment is needed, continue to slowly turn the manual depth screw. Stop when the desired tow-point position has been achieved.

At times when paving with automatic grade control, the tow-point cylinder may run out of adjustment. Use the aforementioned technique to alter the tow-point height or to adjust the line of pull while paving.



User Tip: *It is virtually impossible to "gain back" cylinder stroke when using an averaging ski or if a single sensor is installed close to the tow-point connection. Remember, any changes in screed elevation occur over a long distance when using an averaging ski or when the sensor is close to the tow-point connection. Normally, the tow-point moves very small distances when a ski is used and manual tow-point adjustment isn't required.*

[CALIBRATING TOW-POINT CYLINDER VALVES]

No matter what type of grade and slope control system is being used, the tow-point cylinder valves must be calibrated on each side of the paver. Calibrating the tow-point cylinder valves assures that the tow-points move up and down smoothly and in the correct timeframe in response to signals from the grade and slope control module.

Each tow-point cylinder valve is calibrated independently. The calibration data for each tow-point cylinder valve is stored in the control module of the Cat Grade Control System. On other systems, data for the left tow-point cylinder valve is stored in the left control box. Data for the right tow-point cylinder valve is stored in the right control box. Clearly mark the right and left control boxes, so they are always installed on the correct side of the screed.

Most modern pavers, have proportional tow-point cylinder valves. When calibrating a proportional tow-point cylinder valve, set the amount of current that is necessary to move the valve spool enough to send a small amount of hydraulic oil to the head end or rod end of the tow-point cylinder. It should move a small distance and stop in a short period of time.

If the tow-point cylinder moves too far or too fast, the screed will create roughness in the bituminous layer. If the tow-point cylinder takes too long to respond, the screed will not place the correct thickness.

Note: *Instructions for calibration of tow-point cylinder valves when using the Cat Grade System are in the Operation and Maintenance Manual. Always follow the instructions for the paver model and the type of grade and slope control being utilized.*

To perform tow-point valve calibration, start with the display in the Grade Control mode. Select the menu icon on the touch-screen display. Then, select the Installation bar. There are separate bars for left and right valve calibration. Select the side to be calibrated.



Display showing the start of calibration for the left tow-point valve.

The Cat Grade System uses an "Auto Valve Calibration" process. Be sure to select either the right or left valve and calibrate one side of the paver at a time. Enter the Installation menu and follow the prompts that are shown on the display.



Display with instruction to center the tow-point.

The prompts on the screen provide instructions for completing tasks such as centering the tow-points and setting the upper and lower limits of tow-point travel. Once all of the manual requirements for tow-point valve calibration are completed, the system will perform an automatic self-calibration. When the calibration has been successfully completed, a message on the display indicates it's time to depress the button adjacent to the OK icon.

When calibrating the tow-point cylinder valves for any type of grade and slope condition, follow these guidelines to assure accurate calibration.

- Hydraulic oil at normal operating temperature
- Paver parked on a flat and level surface
- Tow-points centered
- Screed in float mode
- Slack removed from the tow-point connections
- Screed nulled
- Transmission in Pave mode
- Parking brake set
- Propel lever fully forward
- Speed control dial turned up halfway



Display with instruction to set the lower limit of tow-point travel.

Calibration of tow-point cylinder valves is not part of a scheduled maintenance program. Initial calibration of the tow-point valves is required when installing Grade Control systems. Once the system has been calibrated and performance has been verified, recalibration is unnecessary unless:

1. The grade control system is moved to another paver, or
2. The tow-point cylinder or tow-point valve have been replaced.

There may be times when it is suitable to recalibrate the grade control system. Examples are when the paver comes out of storage after sitting idle for a long period of time or before starting a project with a reward or risk associated with ride quality.

[AVERAGING SKIS]

Averaging skis are commonly used on projects where an improvement in the longitudinal profile (smoothness) of the structure is needed. An averaging ski is designed to provide either a physical reference that ignores high spots or low spots in the grade that is being paved or to electronically average three or four measurements taken by grade sensors over a given distance.

Two types of averaging skis are commonly used: mechanical, contact-type skis or electronic, non-contact skis. Each type has its own specific advantages.

Electronic non-contact skis

Electronic, non-contact skis have three or four equally spaced grade sensors along the length of the beam that carries the sensors. Typical beam length is 9 meters (30'). There are also skis with 12-meter (40') beams or longer. Non-contact skis can be set up to read a reference outside of the paving width or inside of the paving width.



Two mounting arms, either single or scissor-type, attached to the tow arm carry the non-contact ski.

It is important for the crew to have the correct mounting arms in order to install a non-contact ski. Two single arms are adequate to carry a ski that will be installed inside the paving width, or outside the paving width if the paving width is relatively narrow—say 3.6 meters (12'). The crew will need longer scissor-type mounting arms in order to install the ski to reference the grade outside the paving width for wider paving—say 4.9 meters (16').



Cat non-contact averaging ski installed to reference grade outside the paving width.

Install the non-contact ski so the center of the beam (balance point) is located somewhere between the center of the tow arm and the tow-point connection. The closer the balance point is to the tow-point connection, the slower the screed reaction will be.

The Cat non-contact ski has three vertically adjustable poles that carry the grade sensors. Measure the height of each sensor above the grade reference. They do not have to be exactly the same height, but they all should be within the recommended range of 40 to 46 cm (16" to 18"). When benchmarking (nulling) the grade control system on that side of the paver, each sensor will be measuring the grade based on its own benchmark.

Follow these installation guidelines for all types of electronic, non-contact skis.

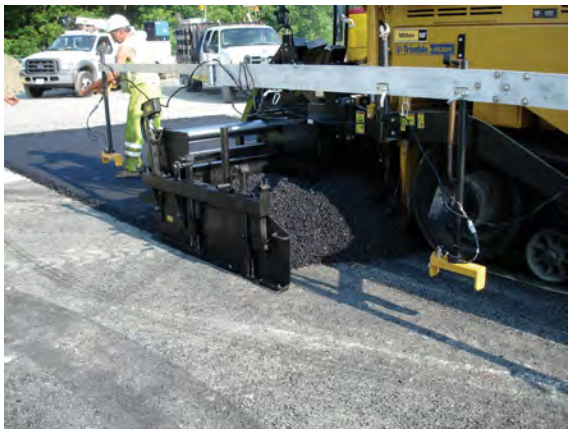
All three grade sensors on the Cat non-contact ski should be parallel to the direction of paving with the cable connection on the sensor toward the rear of the paver.



Cat non-contact ski setup to reference grade inside the paving width.

User Tip: *On some types of non-contact skis, the cable connections that attach to the sensors are numbered. The cable connection labeled as number one should be attached to the sensor closest to the front of the beam. The other cable connections attach to the sensors in numerical order, front to back. If the crew attaches the connections to the sensors in reverse order, the grade control system may act erratically. Caterpillar recommends clearly labeling each cable end, so the crew always installs the cable connections in the correct sequence.*

When a non-contact averaging ski extends over the screed, the last grade sensor will use the surface of the new bituminous layer as the grade reference. This reference will probably provide the most uniform grade measurement of all the sensor signals sent to the control module. Therefore, the averaging factor is improved and the ride quality of the layer will be enhanced. Caterpillar recommends, whenever possible, that the non-contact ski be setup to have the last grade sensor over the new bituminous layer.



The last sensor on the Cat non-contact ski is set to reference the surface of the layer directly behind the screed.

User Tip: *Some sonic grade sensors have foil transducers that emit sound pulses. Foil transducers can be affected by extreme heat and asphalt fumes. When setting up a sonic sensor with a foil transducer over the hot bituminous layer, be sure to inspect that sensor at the end of the shift. Look for wrinkles or the accumulation of an oily film on the transducer. Be prepared to replace the damaged transducer.*

User Tip: *When using any type of averaging ski, Caterpillar recommends taking off from the starting reference in the manual mode. Establish layer thickness manually, then benchmark the ski and enter the Auto mode for grade control. Remember, making depth changes when using an averaging ski takes a long time. It's easier to make depth changes manually and let the ski do the work of improving smoothness.*



Cat non-contact averaging ski with end sections folded and set up to use only the center sensor for a single sensor application.

As pointed out earlier in this unit, the number of sensors that are active on the Cat non-contact ski can be selected by using the Sensor Select function on the display.

When utilizing the Sensor Select function to choose one active sensor on the non-contact averaging ski, the center sensor measures the grade reference and sends a signal to the control module. It is not necessary to remove the ski. It may be necessary to position the mounting arms in a location that allows the center sensor to provide the best screed response. The remaining sensors may need to be disconnected in order to eliminate feedback by the active sensor.

Non-contact averaging skis are preferred for some applications rather than mechanical, contact-type averaging skis. Some of the reasons may include:

- Time constraints. Since the non-contact ski is attached to the side of the paver and is independent of the grade, it remains on the paver when the crew relocates to a new position on the project. Saving time is especially critical during night shifts when there is a need to reopen a roadway to traffic at a certain time.
- Obstacles in the grade. On some projects, obstacles may be encountered like drain inlets or utility boxes. Contact-type skis are affected by these obstacles. Non-contact skis can pass over these obstacles easily.
- Grade reference is acceptable. Electronic averaging skis do not have as much averaging capability as mechanical skis. However, if the grade reference (milled surface, for example) does not have extreme roughness, an electronic averaging ski is adequate and will be preferred because of its convenience compared to a mechanical ski.
- Smoothness specification is moderate. There are many smoothness measurement systems. Profile Index, which was the most common system for many years, is being replaced in many areas by the International Roughness Index. Other areas use a system referred to as Ride Number. Each Public Works Department chooses the system it prefers and then sets standards for the contractor to meet. If those standards are somewhat easy to satisfy, then an electronic averaging ski will be preferred because of its set-up convenience.

Mechanical, contact-type skis

Mechanical averaging skis have been in use since at least the 1950s. Their development was based on the need to improve ride quality even more than what can be accomplished by the floating screed. Remember, the floating screed tends to minimize grade deviations based on the distance from the tow-point connection to the front of the main screed. A mechanical averaging ski doubles or triples that length. A mechanical ski also tends to isolate the tractor portion of the paver from grade deviations.

Numerous references are used with mechanical skis. Some of the first were traveling stringlines and solid beams. Those types of averaging devises provided a grade reference for a sensor that was a 50 percent improvement. In other words, if there is a 25 mm (1") bump in the grade, that deviation will be 12.5 mm (0.5") at the center of the averaging device.

Modern mechanical skis provide a reduction of grade deviations that is approximately 88%. In other words, if there is a 25 mm (1") bump in the grade, that deviation will be 3 mm (0.125") at the center of the averaging device.

Like the electronic, non-contact skis, typical mechanical skis are 9 m (30') long. Longer 12 m (40') versions are also available.

Two basic types of mechanical skis are used: skis that ride outside the paving width and skis that ride inside the paving width.

The Cat 9-meter (30') multi-articulated outboard leveler reads a grade reference outside the paving width. This mechanical averaging ski is also available in the 12 m (40') length. The 9-meter (30') outboard leveler consists of three 2.5 m (8') aluminum sections that are bolted together after transportation to the worksite. Each section carries two shorter beams (about 1 meter / 3 feet long) through articulated connections. Each short beam carries two short contact runners (about 30 cm / 12" long), also through articulated connections.

The 9 meter (30') ski will bridge, or fill in, a depression up to 9 m (30') long. The 12 meter (40') version extends that bridging capability

The outboard leveler is extremely good at reducing roughness in the grade being referenced. When a contact runner goes over a bump, that bump is reduced by 50 percent at the articulation connection between the contact runner and the short beam. The deviation undergoes another 50 percent reduction at the articulation connection between the short beam and the main beam section. Another 50 percent reduction of the remainder of the deviation occurs at the center of the leveler. The three reductions in roughness represent an 88 percent improvement at the center of the leveler where the grade sensor should be located.



Cat 9-meter (30') multi-articulated outboard leveler averaging adjacent layer, while matching the longitudinal joint.